

Is cartoonized life-vlogging the key to increasing adoption of activity-oriented wearable camera systems?

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ABSTRACT

Health science researchers studying human behavior rely on wearable cameras to visually confirm behaviors in real-world settings. However, privacy concerns significantly impede their adoption. Lens orientation and activity-oriented cameras have potential in balancing the need to visually validate the wearers' activities while reducing privacy concerns. To increase adoption and further alleviate privacy concerns while maintaining utility, generative stylizing approaches, like cartooning using generative adversarial networks (GANs), have recently shown promise. We investigate different cartoon-based obfuscation of activity-oriented footage through two studies. The first deploys crowdsourcing methods ($n=60$), while the second is experiential, where participants ($n=49$) don the device for an entire day and report concerns on their footage. Our findings support that cartoonization of activity-oriented data significantly reduces privacy concerns, particularly among bystanders in high privacy-concerning scenarios, while maintaining context verification (90% of participants). Through thematic analysis, we provide further insight for the community on best practices for cartoonization of activity-oriented videos.

CCS CONCEPTS

• **Human-centered computing** → **Empirical studies in HCI**; • **Information systems** → **Crowdsourcing**; • **Computing methodologies** → **Scene understanding**.

KEYWORDS

privacy concerns, activity-oriented wearable cameras, obfuscation techniques, generative adversarial networks, qualtrics survey

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1 INTRODUCTION

Embedded wearable cameras are becoming more ubiquitous among researchers interested in studying human behavior in real-world settings, allowing them to visually confirm problematic health behaviors. Wearable cameras provide accurate and impartial data on the normal daily activities of the user (i.e., wearer) and their interactions with the environment [5, 7, 14]. Prominent examples of this in the research community include SenseCam [7] for lifelogging and ActiSight [3] for detection of hand-related activities (e.g., smoking, eating, and drinking). One of the major challenges in the study of human behavior is the capture of authentic or naturally occurring behavior, and given the known privacy concerns associated with wearable cameras [4, 15, 16, 23, 27], researchers have attempted to design methods that reduce privacy concerns while maintaining their utility for visually confirming human behavior. One type of camera that has generated concerns is the egocentric camera that captures first-person views. Alharbi et al. [4] set out to study cameras on different body locations with different lens orientations to provide device recording affordances that reduce surveillance and social discomfort compared with egocentric cameras. These are referred to as activity-oriented cameras (AOCs) because they are designed to capture a particular activity rather than the field of view of the wearer. To capture hand-related activities of the wearer (e.g., eating, smoking, drinking), Alharbi et al. showed through multiple studies [3–5] that the use of a wearable video camera in a necklace form factor, with the lens oriented upward toward the mouth, combined with the use of a thermal sensor that aids the obfuscation of RGB video by distinguishing the foreground pixels (i.e., the wearer and activity of interest) and applies obfuscation methods to the background pixels, significantly increases people's willingness to use wearable cameras. AOCs have thus been shown to be highly

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promising for the capture of hand-related behaviors; however, the obfuscation methods tested are quite primitive, including applying complete masks (the most extreme obfuscation method in which background pixels are set to a single color, such as black), blurring (which retains some information about the colors and shapes of the objects), or canny edge (which maintains the shape of the objects by only displaying an outline). Despite these advances, the partial nature of obfuscation makes the wearer's face identifiable, and participants still report privacy concerns for bystanders that are recorded without consent.

Researchers evaluated blur and edge filters on footage from activity-oriented wearable cameras and found the blur filter to be most preferred by the wearers [5]. However, several references [6, 9, 10, 13, 19] have shown that using cartoons/avatars for obfuscation is more privacy preserving over image-processing filters such as blurring, canny edge detection, and pixelation. Cartoon-based obfuscation techniques are being used to deidentify faces [6, 10, 11, 17, 20, 21, 26], entire persons [6, 19], and the surrounding environment [10, 13] to preserve user privacy in surveillance videos. Cartooning can be implemented in different ways, from implementing a clever combination of canny edges and box blurs to using advanced generative algorithms [8, 20, 25, 28, 30]. The latest developments in generative deep-learning techniques have led to the creation of high-quality cartoonization models, such as Cartoon-GAN [8] and white-box [28] models that convert real-world footage into anime-style cartoon videos. However, the influence of generative cartooning methods on privacy concerns and user acceptance using AOC systems has not yet been studied.

Videos recorded with an AOC contain three important elements that can pose privacy concerns to the wearer and bystanders. These include (1) the wearer's face, which roughly occupies one-third of the frame [4, 5]; (2) bystanders captured in the field of view of the camera; and (3) the surrounding environment of the wearer. In order to maximize privacy, we tested a combination of cartooning techniques that address privacy issues across all three elements of the scene. Our approach uses an avatar face to obfuscate the wearer's and bystanders' faces; the bystanders' bodies and surrounding environment are obfuscated using a white-box Cartoon-GAN [28] for an enhanced cartoonized effect. In addition, this method also helps obfuscate information on screens, phones, or other sensitive personal objects in the user's environment.

In this paper, we aim to show the benefits of cartoonization of video in AOCs in reducing privacy concerns. We set out to answer the following questions: (R1) How does cartoon-based obfuscation of activity-oriented video footage affect the wearer's and bystanders' privacy concerns and visual confirmation utility? (R2) Do wearers prefer cartoonization using an avatar over other obfuscation methods in AOC video footage? The following contributions in this paper enable us to answer the above research questions: (1) a unique cartooning approach to reduce privacy concerns of the wearer and bystanders in the field of view; (2) a Qualtrics survey with 60 participants to measure the privacy concerns of participants deployed using Prolific (a higher standard of online research); and (3) an in-person study with 49 participants for an entire day, where participants use the device and perform normal daily activities, followed by 1:1 semi-structured interviews with the participants to understand their privacy concerns.

Our results show that our cartoon-based obfuscation approach of AOC video significantly reduces privacy concern levels, particularly for bystanders in high-risk scenarios, all while maintaining the ability to visually confirm context. By increasing acceptance of wearable video cameras, our research can drive greater adoption of AOC and ultimately improve understanding of naturally occurring problematic human behavior in real-world settings.

2 RELATED WORKS

Prior wearable cameras, such as Sensecam, were used as memory recall tools [14] and as behavioral monitoring tools to capture eating patterns [7]. However, privacy concerns of Sensecam users and bystanders remained unclear. To test the feasibility of using privacy filters for embedded cameras, Erdelyi et al. [10] developed a video cartooning approach combining blurring and edge detection. Their assessment pointed to the balance between privacy and utility compared with other approaches, such as blurring and pixelation. Similarly, Winkler et al. [29] proposed a mean shift filtering method to preserve privacy by globally cartoonizing every frame. However, the impact on user acceptance of the video was unknown, as was the utility of the video footage in maintaining visual confirmation of the wearer's activity. More recently, researchers have evaluated people's perceptions of different wearable camera affordances and coined a new, activity-oriented camera (AOC) [4]. To reduce privacy concerns, the findings suggested wearable cameras should orient the lens toward the activity of interest [4] while using thermal data to identify the foreground pixels (i.e., the wearer and objects in close proximity) and obfuscate all other background pixels (partial obfuscation) [5]. Through a subsequent study, the authors concluded that AOCs using blur and edge filters achieved a balance between privacy and visual confirmation of the wearer's activity [5]. However, filters such as blur and edge allow for identifying features such as the texture and color of clothes and personal belongings of both the wearer and bystander [6]. Avatars and generative methods, on the other hand, have been shown to more successfully obfuscate the wearer and bystander while maintaining the ability to visually confirm context [19]. This motivates our first research question regarding assessing whether cartoon-based obfuscation of activity-oriented video footage affects wearers' and bystanders' privacy concerns and/or visual confirmation utility.

Based on data collected from an AOC, video can be split and obfuscation applied to each of three segments in a scene: (1) the wearer's face and hands; (2) bystanders; and (3) the environment (everything else in the scene). Researchers have proposed several techniques for face obfuscation, for example, convolution-based filtering [22], geometry-based face morphing [17], and in-painting [22], as well as deep learning methods such as the use of generative adversarial networks (GANs) to distort faces [24]. GAN-based obfuscation can also be applied to the face and body [6]. Moreover, Hasan et al.'s [12] cartoon obfuscation algorithm for deidentifying sensitive objects and personal details in the environment showed that the cartoon approach strikes an acceptable balance between understanding the video's context and protecting privacy. GANs can also be utilized to obscure the wearer's surrounding environment. CartoonGAN [8] is one of the more popular GANs for cartooning



Figure 1: (a) Study design for the survey. (b) Illustration of a person wearing a necklace form-factor activity-oriented camera with lens oriented upward. The dashed lines represent the field of view of the camera.

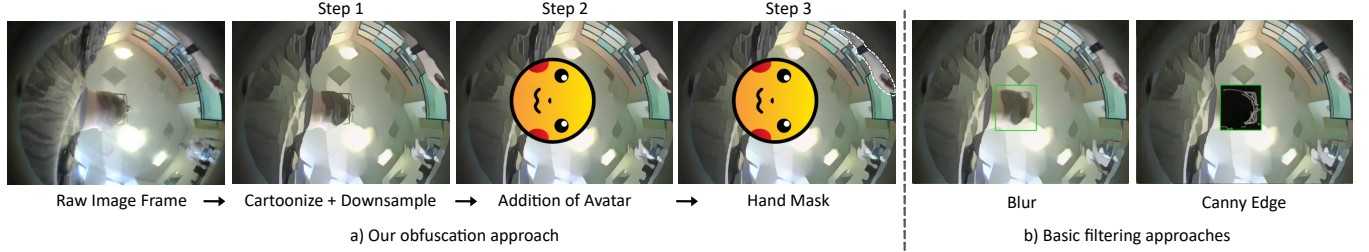


Figure 2: Obfuscation Pipeline: The process of generating a cartoonized obfuscated frame from a raw frame.

real-world videos and has been shown to outperform other cartoonization GANs in quality. Wang et al. [28] improved cartoon quality and overall image quality by using white-box representations (separating out the ability to discriminate between different textures, structures, and surfaces in images to reduce the black-box nature of CartoonGAN). Methods such as blurring and blocking [18], as well as masking, silhouette, avatar, and pixelation, have been tested by Li et al. in their Amazon Mechanical Turk (MTurk) survey [19], where they highlight users’ preference to replace people with human avatars to address privacy concerns.

The above-mentioned obfuscation methods were designed to be used on egocentric video footage, focusing primarily on people present directly in front of the camera, and because the wearer is not in the scene, they fail to assess the effects of obfuscation on the wearer. Moreover, these approaches have not been tested on footage from AOCs, especially in studies involving normal daily activities. This motivates the need for our second research question, namely whether wearers prefer cartoonization using an avatar over other obfuscation methods in AOC video footage. In this paper, we investigate these two research questions and perform thematic analysis to identify themes that optimize the design of cartoon-based obfuscation.

3 METHODS

To answer our research questions, through two independent studies, we evaluated the privacy concerns of wearers and bystanders in AOC obfuscated video footage and the ability to verify their context. Here we describe the first Qualtrics survey with 60 participants and the second in-lab study with 49 participants.

3.1 Qualtrics Survey

We conducted a Qualtrics survey [2] to analyze the impact of cartoon-based obfuscation techniques on the wearer’s and bystanders’ privacy concerns and the ability to verify the video context. Participants for the survey were recruited using Prolific’s Platform [1]. The survey was completed by a total of 60 participants, whose ages spanned from 23 to 76 years old (mean [SD], 41.6 [14.0] years). Overall, 49 individuals identified as White, 2 as Black/African American, 6 as Asian, and 3 as Hispanic/Latino; 30 participants identified as female, 29 as male, and 1 as nonbinary. A small number of the participants (8.3%), had worn a camera, such as a Go-Pro, in the past, whereas the majority of the participants had not used any form of wearable camera. At the start of the study, participants were asked about their background and a set of baseline concern-level questions. The participants were randomly assigned to either watch videos processed using our cartoon-based obfuscation approach (cartoon condition) or to watch raw videos (raw condition), as seen in Figure 1. Sixty participants were recruited and randomly assigned to either the cartoon or raw video footage condition. If assigned the cartoon block, participants watched four videos: (1) low-risk scenario with the wearer obfuscated; (2) low-risk scenario with the bystander obfuscated; (3) high-risk scenario with the wearer obfuscated; and (4) high-risk scenario with bystander obfuscated. However, if assigned the raw block, participants would see the same four videos without any obfuscation. The order of the four videos within each block was shuffled for all participants to mitigate carry-over effects. After each video, participants were asked to recognize the context of either the wearer or the bystander and then answered a question asking them how concerned they would be if they were in the wearer’s/bystander’s position using a Likert scale (see questionnaire in the appendix).

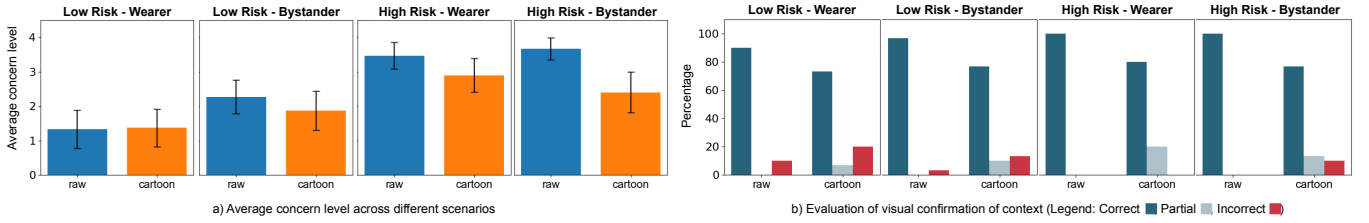


Figure 3: (a) The mean (95% confidence interval) concern level of participants in each of the conditions (raw, cartoon) given low-risk or high-risk scenarios with questions focused on either the wearer or bystander. The y-axis represents the Likert scale (0, not concerned at all; 4, extremely concerned) (b) The percentage of participants that got the context (of the video shown to them) correct, partially correct, or incorrect. Each column represents the scenario described.

Video	Statistic (W)	p-value	Raw Median	Cartoon Median
Low-Risk Wearer	453	0.97	1	1
High-Risk Wearer	544	0.11	4	4
Low-Risk Bystander	516.5	0.32	2	2
High-Risk Bystander	646	0.00079	4	3

Table 1: U-tests (Wilcoxon rank-sum test). Four hypothesis tests are being conducted, therefore, p-values are set to $\alpha = 0.0125$.

We captured the low-risk and high-risk scenario using two actors wearing a necklace form factor-based AOC (lens oriented upward toward the wearer’s face) comprising an RGB sensor equipped with a fisheye lens so that the wearer’s face and hand-related activity were visible in the camera’s field of view [5]. The low-risk concern scenario comprised a wearer working at a desk using their laptop and a bystander entering the scene to drink water from a water bottle at the wearer’s desk. The high-risk concern scenario comprised a wearer using the restroom while a bystander was using an adjacent urinal, after which they proceeded to wash their hands at a wash basin. The wearer’s face and hands and the bystanders’ faces and bodies were annotated using annotation software.

Our cartoon-based obfuscation method consisted of obfuscating the wearer, the bystander, and the surrounding environment using a three-step process as shown in Figure 2. (1) We first globally cartoonized the entire frame using the white-box cartoonization GAN [28] and down-sampled the image. (2) We then superimposed faces in the frame with avatars. Subsequently, (3) we used the labels for the wearer’s hand and the raw image to make the hands and objects in hand visible (in the final frame) so that the wearer’s activities were not obscured. Our goal was to preserve the wearer’s activity context as much as possible while addressing the privacy concerns of both wearers and bystanders. This led to choosing avatars for obfuscating faces to maximize privacy while keeping the wearer’s hands unobfuscated to allow for context verification.

In addition, at the end of the study, we showed the participants blurred and canny-edged versions of the videos (Figure 2b), where the faces of wearers and bystanders were blurred/edged instead of blocked using avatars. We then asked them to compare and rank our obfuscation methods with blur and edge filtering based on their willingness to wear the camera, given that the footage was obfuscated using one of these methods.

3.2 In-Lab Study

Although effective at providing an initial understanding of privacy concerns, the online Qualtrics survey did not afford participants the opportunity to utilize the device or review their own data. To understand potential differences in participants’ privacy concerns when directly interacting with an activity-oriented wearable camera and viewing their own footage, we conducted an in-lab study ($n=49$) in which participants engaged in their normal daily activities while wearing the AOC in a controlled social environment. Throughout the study, participants were provided with breakfast, lunch, dinner, and snacks; they were also provided with video games and asked to perform the following activities: (1) eat; (2) rest; (3) work on their laptops or phones; (4) use the restroom; (5) play board games; and (6) watch television. Additionally, participants were permitted to leave the laboratory and return between scheduled activities. At the end of the study day, the data from an eating episode (involving hand-related activity) collected during the day was obfuscated using blur, edge, and our cartoon-based approach. Semi-structured interviews were conducted to show participants short video clips of the obfuscated data, along with raw footage for comparison, to solicit their feedback on the effectiveness of each obfuscation method. During the interviews, the questions were aimed at understanding the participants’ filter preferences, privacy concerns, and apprehension regarding data sharing in diverse social situations, such as sharing the footage with friends, family, medical professionals, or strangers. Thematic analysis was conducted to assess the participants’ obfuscation preferences. The analysis used a predetermined set of codes based on the researchers’ expectations of the participants’ responses to the questions posed. In instances where the assigned codes did not adequately capture the context of the response, new codes were created, and themes were generated based on the frequency of different codes.

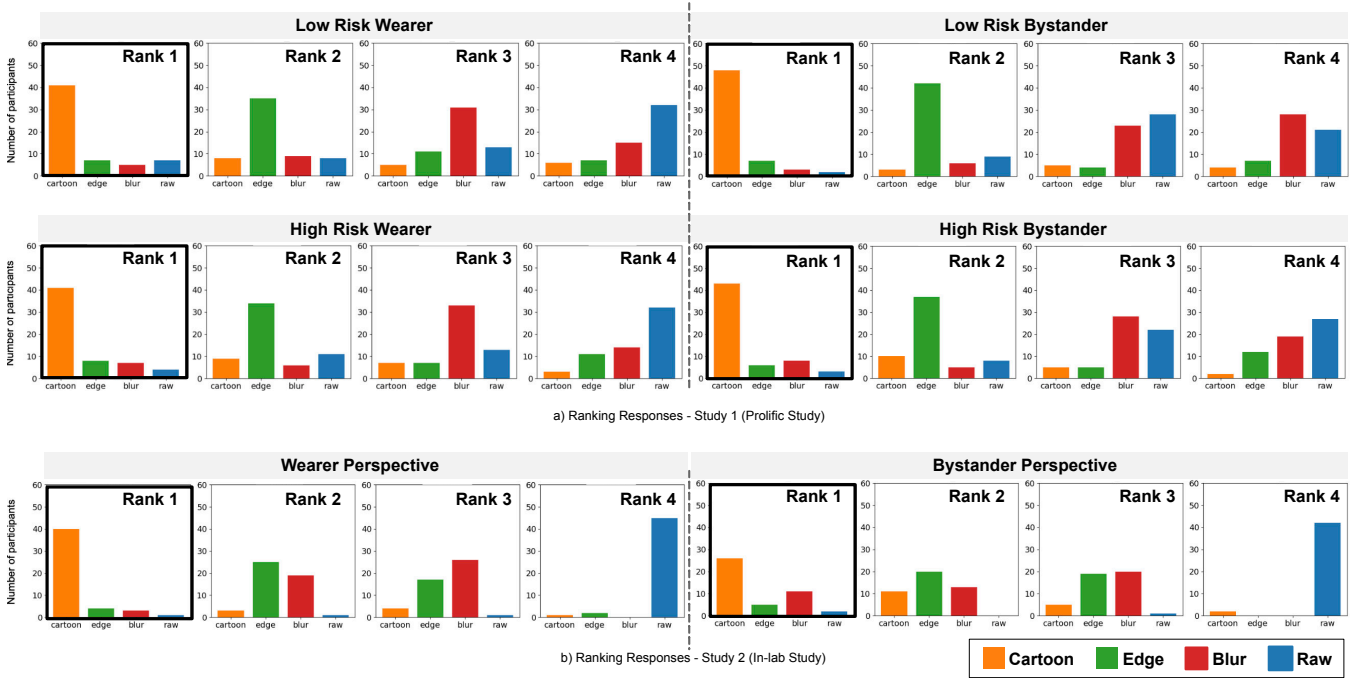


Figure 4: Ranking Question for (a) prolific study and (b) in-lab study. The distribution are displayed for participants' rankings for each of the obfuscation techniques in response to the given video scenarios (range, 1–4).

4 ANALYSIS AND RESULTS

We conduct quantitative analysis on data from our Qualtrics survey, which includes concern levels for raw versus cartoon conditions, context verification, ranking of different obfuscation techniques, and qualitative analysis from themes arising from our interviews with participants in the in-lab study.

4.1 Raw Versus Cartoon

We analyzed participants' responses when asked about their concern level if they were in the same situation as either the wearer or the bystander in the video they were watching. Figure 3 shows the average concern level of participants for each scenario. The increase in concern from low-risk to high-risk videos is more pronounced for wearers than it is for bystanders. This indicates that if an individual were a wearer, low-risk videos were viewed with low risk, and high-risk videos were viewed with high risk. However, if an individual were a bystander, low-risk videos were viewed with moderate risk, and high-risk videos were viewed with high risk. In the low-risk wearer, high-risk wearer, and low-risk bystander videos, there was no statistically significant difference between the raw and cartoon conditions (Wilcoxon rank-sum test [Mann Whitney U Test]; Table 1). However, there was a statistically significant (p -value = 0.00079) difference between raw and cartoon conditions in the high-risk bystander scenario.

4.2 Context Verification

We defined key terms for context in the low-risk and high-risk videos to analyze and tally participants' responses when asked

about the context of the video they watched. If the participants' responses contained all or some of the defined keywords for the scenario in which they were identifying the context, we classified the responses as correct responses. If the participants missed some of the terms, we classified the responses as partially correct; otherwise, we classified their responses as incorrect. As seen in Figure 3b, in the low-risk wearer scenario, 90% of the participants identified context correctly in the raw condition, and about 80% (73.3% correct + 6.7% partially correct) identified the context in the cartoon condition. In the low-risk bystander scenario, 96.7% participants identified the context accurately in the raw condition, and about 86.7% (76.7% + 10%) identified the context in the cartoon condition. In the high-risk wearer scenario, 100% identified the context in both raw and cartoon conditions. In the high-risk bystander scenario, 100% identified the context accurately in the raw condition and about 90% (76.7% + 13.3%) were able to identify the context in the cartoon condition.

4.3 Ranking

When asked to compare the different obfuscation methods at the end of the study, about 72.1% of the participants ranked the cartoon approach with the avatars on faces to be the most preferred over edge, blur, and lastly raw (Figure 4a).

4.4 In-Lab Results

During the semi-structured interviews, when participants were asked to compare the different obfuscation methods at the end of the in-lab study, a majority (71.7%) of the participants ranked the cartoon approach with the avatars on faces to be the most preferred

Theme	Description
Biometric identifiers	Our obfuscation approach was the most privacy preserving because it curbed recognizable personal traits
Wearer's avatar	The avatar on the wearer's face, although most privacy preserving, obscured certain finer details during certain activities
Bystander avatar	This approach is privacy preserving but can draw unwanted attention at times

Table 2: Recurring themes captured during the in-lab semi-structured interviews

over edge, blur, and lastly raw (Figure 4b). Through thematic analysis of the semi-structured interviews, three main themes were revealed, which are presented in Table 2.

4.4.1 Biometric identifiers. A total of 53.1% of participants reported that our cartoon-based obfuscation technique obscured their facial features and mitigated identifiable biometric markers and personal traits. When asked to rank the obfuscation methods based on their preference, participant 38 (P38) stated (referring to our obfuscation method as "cartoon"): *"Cartoon best because none of my features are distinguishable not even the shape of my head. Blur #2 because you can see the shape of my head but can not distinguish any of my features. Edge #3 because you can make out my features, but not recognize me. Raw #4 because you can easily recognize me."* The participant appreciated that our cartoon-based obfuscation approach was successful in occluding several personality traits. P25 stated, *"Cartoon completely occludes any recognizable features; skin color, facial features. Blur only gives very rough contours and blurred colors. Edge gives only contours, but facial angles might be identifiable or informative. Raw shows all information, so it is the least. Maybe if I had a prominent chin, you could recognize it with edge. With blur, I think the only way to get caught is by a birthmark or tattoo or something."*

4.4.2 Wearer's avatar. A total of 4.1% of participants reported that our cartoon-based obfuscation approach hindered the identification of finer details during certain activities. P44 stated, *"Cartoon is totally blocked out, you wouldn't know what you are looking at. Edge still has outlines. Those outlines don't show any personal features, and so it's better than a blur for that reason."* Similarly, P47 stated, *"Cartoon is something covering your face completely, so there is nothing to analyze besides your hands or whatever is showing. Edge has no color or anything that could be anyone's face. Blur is still not an accurate showing of the face, so it is still somewhat privacy protecting vs raw."*

4.4.3 Bystander avatar. A total of 12.2% of participants noted that cartoon-based obfuscation preserved the privacy of bystanders while being fun; P28 stated, *"[The avatar in cartoon-based obfuscation approach] removes their face, and is funny."* Despite the presence of avatars obfuscating bystanders, some participants were unable to identify them in the footage (likely due to the low number of pixels representing the bystander). On the contrary, P46 expressed that *"Cartoon draws attention to the person in the background. I wouldn't have even noticed him in the raw, and he blends in in the blur and edge versions."* P45 stated, *"The cartoon draws your attention. Edge is also noticeable compared to blur."* P39 also mentioned that, *"The edge kind of defined the face a little bit. Cartoon draws attention that there is someone there."*

5 DISCUSSION

Our results show that our cartoon-based obfuscation approach can alleviate not only wearers' but also bystanders' privacy concerns; for especially high-risk scenarios (e.g., using the restroom), there is a significant improvement in privacy concern level when participants view footage in the cartoon condition compared with raw video footage. During the use of such AOC systems for studying behavior, researchers aim to study the wearer's and bystander's behaviors (e.g., working at a table, eating); therefore, it is essential to maintain the ability to identify the activities even after the obfuscation process. Our results further show that the cartoon-based obfuscation strikes a balance between curbing privacy concerns and maintaining the ability to visually confirm context within the scene. Results from our Qualtrics survey show that despite the use of intense image processing using a GAN for cartoonization and down-sampling the frame thereafter, 90% of the participants assigned to the cartoon block were able to accurately determine the context in the video. In addition, our cartoon-based obfuscation approach was preferred by users not only in the Qualtrics survey but also by participants in the in-lab study. Our findings in the Qualtrics survey also hold true in the in-lab study, where participants donned the device and were able to visually see their own data.

The themes generated from the analysis showed that participants ranked our cartoon-based obfuscation approach the best because they felt it was the most privacy-preserving approach, primarily because identifying features (e.g., facial features, skin color) of the wearer were more difficult to recognize. Although privacy preserving, participants noted that using an avatar in the cartoon-based obfuscation method resulted in complete occlusion of the wearer's face, impeding the ability to identify nuanced behavioral cues, especially in activities such as eating. Based on participants' responses, we conclude that even though using an avatar might be the most privacy preserving for the wearer, determining fine-grained activities might be more difficult. To achieve a balance between visual confirmation and privacy in instances where fine-grained activity recognition such as eating or chewing gestures is important, we could use the edge or blur filter. However, with more advances in artificial intelligence, we could map an avatar's facial features to that of the wearer's face to mimic exact fine-grained facial activities. Participants stated that the addition of the avatar to the bystander's face successfully reduced their wearer and bystander privacy concerns. However, they were drawn to the presence of a bystander without being aware of the bystander's presence had the avatar been absent. Thus, unless this is desired, to alleviate the privacy concerns of bystanders, it may be advisable to refrain from using avatars on individuals positioned that are difficult to identify, as

well as those positioned at a considerable distance from the wearer, as this may attract undue attention. Although the GAN and down-sampling process utilized in our obfuscation approach inherently made it challenging to recognize bystanders, the participants' feedback regarding the effectiveness of avatars in obfuscating biometric identifiers warrants consideration. Thus, it may be advantageous to incorporate avatars in situations where bystanders are in closer proximity to the wearer.

6 CONCLUSION AND FUTURE WORK

This paper presents a framework for cartoon-based obfuscation and evaluates its effectiveness in mitigating privacy concerns for both wearers and bystanders in the scene while maintaining the ability to identify video context. A Qualtrics survey and an in-lab study were conducted to gather participants' perceptions of privacy concerns and to evaluate their ability to identify the context in AOC videos. Our findings indicate that our cartoon-based obfuscation approach significantly reduced privacy concerns, especially for bystanders in high-risk scenarios, without impeding the ability to discern the context of the video across scenarios. Although avatars applied to both wearers and bystanders effectively preserved privacy, the use of avatars on bystanders situated at a substantial distance from the wearer could draw unwanted attention. Additionally, using avatars to obfuscate the wearer's face could impede the validation of fine-grained activity. Our future plans include developing techniques to map an avatar's facial features to that of the user to preserve visual confirmation of fine-grained wearer activity. Furthermore, to gauge the effectiveness of these obfuscation methods in real-world settings and to gain a better understanding of naturally occurring behavior, we plan to conduct a free-living longitudinal study wherein participants wear AOCs throughout their day.

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